

The Receipt: An Auditable, Correctable Recommendation for Pantry-Aware Meal Planning

Project 1b — Epicourier-Web (Group 12)

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Abstract

Epicourier prints a recipe “match percentage” in bold that is authored by a language model (Google Gemini) and that *no line of the codebase recomputes* from the expiry dates already in the pantry. We propose **the receipt**: move the ranking out of the model into a pure, published integer function and print, beneath the number, every pantry item, its date, the points it contributed, and the running total — a receipt the user can correct and watch the order move. We surveyed the market with **four LLMs used as independent analysts** (Claude Opus 5 and DeepSeek V4 with web retrieval; OpenAI Codex; and Qwen2.5:32B run locally without retrieval). Of four candidate gaps, **two are dead** and **two were narrowed by our own prompts**; the surviving gap is a four-way conjunction, not a moat. Our four-person team will ship this as three user-facing features — cross-device parity, a pantry-aware link from recommendation to shopping list, and a training-day diet target — unified by the receipt; the market places the first two as table stakes and only the audited combination as a differentiator. The proposal is testable offline with no human subjects. This report is written to be graded on caught errors: it records 129+ corrections, one withdrawn citation of our own, and the cross-model disagreements that produced them.

1 Introduction

Epicourier-Web is a full-stack meal planner (Next.js / TypeScript front end; FastAPI / Python back end; Supabase/PostgreSQL; Google Gemini 2.5 Flash for recommendation). Its twenty implemented use cases span recipe browsing with an “inventory match percentage,” AI meal-plan recommendation, calendar scheduling and completion, a nutrient dashboard with goals and CSV export, gamified challenges, pantry inventory with expiry colour-coding, expiry-prioritised recipe suggestions, and a shopping-list-to-inventory transfer flow.

The extension has a *hard budget*: four graduate students, one month, ≈ 160 person-hours, to build **and test**. Everything below is scoped to that. The **challenge (why)**: the recommendation the product sells hardest is unaccountable. The **thing we build (what)**: a deterministic, user-correctable

receipt behind every ranking. The **benefit (so what)**: the recommendation becomes inspectable rather than trusted on an “AI” label, and we can prove the arithmetic is right offline, exhaustively, in seconds.

2 D1 — Market Survey

2.1 The rivals

We verified **14 live rivals** and **4 confirmed-dead** products from 48 distinct candidates (after name normalisation), each name carrying a live URL a second agent re-fetched. Table 1 is the load-bearing subset.

2.2 Four gaps, interrogated to destruction

We pre-committed five confirming and five refuting items per gap and marked each FOUND/NOT-FOUND after looking. Two gaps died:

- **G2** “nothing measures whether the plan reduced waste” — **dead**. Five shipping products claim it, *and* the measurement is out of budget: the nearest published trial ran six students for a month and found no change [1].
- **G4** “nothing connects a calorie target to this week’s groceries” — **dead**. Eat This Much ships the whole chain; others ship per-training-day macros.
- **G3** “nobody treats the pantry as uncertain” — **narrowed**. Three products do; what survives is that none propagates that uncertainty *into the ranking*.
- **G1** “nobody explains *why* it recommended this” — **narrowed twice in one day**. RecipeFix ships substitution reasoning; Cooklist advertises expiry-aware prompts (on a mockup). What survives is a **four-way conjunction**: no product joins the *named item*, its *actual expiry date*, the *nutrient constraint*, and the *reason for a substitution* into one object a person can read and correct.

We state on the poster that this is a *conjunction gap*, *not a moat*.

2.3 The two-model rule

A rival “counts” only if two of our LLMs name it, *or* one supplies a live URL. Table 2 applies it honestly: six rivals clear

Table 1: Representative verified rivals (checked 2026-09-13). “–” = not established on the pages we read; not proof of absence.

Product	Who uses it	Main strength	Main weakness / note
Samsung Food+ [14]	All-in-one home cooks	Food List prioritises items nearing use-by	Food+ price unknown; a pillar Claude could not reach (HTTP 403), Codex could
Eat This Much [7]	Dieters wanting auto plans	Nutrient targets + pantry priority + grocery list	Prioritises <i>leftovers</i> by assumed consumption, not shelf life (disagreement D3)
Paprika 3 [11] SuperCook Cooklist [5]	Recipe collectors Ingredient-first cooks Grocery-synced households	Import, meal calendar, pantry, lists Discovery from current ingredients 11,297 iOS ratings; “parsley is 7 days old” suggestion	No verified outcome/ranking rationale No update since 2022-06-15 That screenshot is a Sketch mockup , not a device capture — caught by opening the image
Grocy [8] (OSS)	Self-hosters	Publishes an integer “due score” formula — auditable ranking already ships	Self-host only; not consumer-friendly
RecipeFix [12]	Substitution seekers	“No black-box AI — every substitution comes with the reasoning”	Kills the <i>broad</i> form of our gap
Mealime	Busy households	Simple personalised plans	Discontinued 2026-10-21 (named by both models; caught as stale in Codex’s table)

the two-model bar outright, thirteen more clear the live-URL bar, and three are struck as withdrawn. The three products added last (Cooklist, Remy, Grocy) mattered most — an argument against capping verification by relevance ranking, because our relevance ranking was wrong.

Table 2: Two-model-rule survival (excerpt). ✓=named independently.

Rival	Cdx	Cld	Survives by
Samsung Food+	✓	✓	two models + URL
Eat This Much	✓	✓	two models + URL
Paprika, SideChef, SuperCook	✓	✓	two models + URL
Mealime	✓	✓	two models — <i>but discontinued</i>
Cooklist, Grocy, Remy	–	✓	live URL (added last; highest value)
Kitche, Fridgely, CozZo	–	✓	struck — confirmed withdrawn

2.4 The three features we will build, against the market

Our Project 2 plan is three user-facing features. The market decides how each is positioned (Table 3); our own P13–P20 market-support study places two of the three as *table stakes* and only their audited combination as a differentiator.

- **Recommendation → shopping list.** When a recommended recipe is accepted, compute required quantities, subtract confirmed pantry stock, consolidate duplicates, and add *only the missing delta* to the To-Buy list after a preview the user confirms. Samsung Food (“turn any recipe... into a smart shopping list with one click” [13]) and Mealime [9] already ship the broad workflow; our differentiator is the pantry-aware missing-item delta with provenance — the same line items the receipt already computes.

- **Cross-device / platform adaptation.** The accepted meal and list are usable on desktop and narrow-mobile, state syncs across sessions, and any absent wearable/appliance capability degrades to manual input without dropping the plan. This is a *quality requirement*, not the product gap: incumbents ship multi-platform continuity, so we claim it as parity. Native apps, offline conflict resolution and smart-appliance control are out of a one-month budget.
- **Diet + gym target.** Epicourier accepts a *user-confirmed* daily energy/macro target, labelled with its source and time, ranks recipes against the remaining target and pantry state, and explains the trade-off. It does **not** estimate workout burn and never states medical advice. The calorie-to-grocery chain is already shipped by MyFitnessPal and Cronometer [6, 10] — which is why gap G4 is dead; the wedge is the *auditable* training-day bridge to an explained, pantry-aware meal.

Table 3: The team’s three features, classified against the market (P13–P20).

Feature	Demand	Satur.	Our claim	honest
Recipe → To-Buy list	Strong	High	Table stakes; differ on missing-item delta	
Cross-device adaptation	Moderate	High	Parity / quality requirement	
Diet + gym target	Strong	High (safety-sens.)	Wedge only as audited target→meal→list bridge	

3 D2 — Mission and Stakeholders

3.1 Mission statement

*Every meal planner we verified tells you **that** a recipe matches; none assembles the whole argument in one place.* Epicourier’s own match percentage (RecipeRecommendationModal.tsx:304)

is `match_score (inventory_recommender.py:51)`, a number Gemini writes in prose that no code recomputes. We will move the ranking into a pure Python function scored by a *published integer table* — Grocy’s due score [8], cited in a code comment, with its “20 points per expired ingredient” term **set to zero** because we will not rank food we have labelled expired into a meal — and print the receipt beneath the number. Gemini keeps its job (describing a ranking it was handed) and loses its authority; the user can correct any line and watch the order move.

3.2 The measurable claim (M0), isolated

Table 4: The one measurable claim: metric, threshold, baseline.

Metric 1	Displayed total = sum of displayed line items
Threshold	500/500 recommendations (50 recipes × 10 seeded pantries)
Metric 2	Ranking order stable under permutation of inventory input
Threshold	Byte-identical across 20 shuffles
Baseline	Same harness on today’s build. Today’s API caps at 10 recs/pantry (<code>num_recipes le=10</code>), so the honest baseline is 0 of 300 decomposable scores (30 live calls, 100 cells, 3 repeats), <i>not</i> 0/500.
Cost	Runs offline in seconds; no users, no network.

3.3 Stakeholders (excerpt)

Beyond customer/staff/admin, 18 stakeholders were named, each with a fear and a *testable* design decision (Table 5).

Table 5: Stakeholders (excerpt): each fear maps to a test.

Stakeholder	Fear	Winning decision
Household member with an allergy	An allergen is ranked in	Per-person constraints; no “medically safe” guarantee
Accessibility user	Expiry shown by colour only	WCAG 2.2 AA: second non-colour channel [17]
Security reviewer	Cross-user data access	Fix ownership checks first; RLS threat model [2, 15]
Recipe author	Uncredited copying	Preserve source URL and attribution
Course marker	Claims without runs	Versioned prompts, transcripts, caught-error log
Next semester’s team	Brittle handover	One-command env; claim-to-test register

4 D3 — Milestones and Support Material

4.1 Before / Now / Future

BEFORE (P1a). We authored and ran functional tests (Web 31/32, Backend 18/18), an adversarial pass (69 cases), triaged 18 failures to 17 distinct defects, and *corrected our own findings* — downgrading our headline IDOR to a probable false positive and finding the real defect: the transfer

route returns `success:true` without checking affected rows (silent data loss).

NOW (Project 2), the receipt. Budget verdict first: the NOW plan totals **100.75 h against 100 h** of capacity (160 h – 60 h report/poster/demo); dropping N5b (share-route hardening, 8.75 h) restores an ≈ 8 h margin — an *open team decision*, not a settled one.

- **N1** Claim freeze, one-command env, harness skeleton (12h).
- **N2** Deterministic scorer as a pure, property-tested module (24h) — ports `inventory_recommender.py:159–162` prose into arithmetic; corpus is exactly 50 recipes × 448 ingredients (enumerable).
- **N3** Receipt on screen + authenticated server-side ranking (22h).
- **N4** M0 harness, recorded baseline, correctability measure (24h).
- **N5** Security/data-integrity floor: quantity floor, restock-date merge, affected-rows check (N5a, 10 h); share-route hardening (N5b, 8.75 h, cut candidate).

FUTURE (Project 3). Move food safety from prose into an enforced filter; add per-purchase *lots* (today’s schema sums quantities under a UNIQUE constraint, so “lot” is not representable); uncertainty-aware ranking (*hypothesis* — 851 Reddit entries + ≈ 500 reviews returned zero demand-side hits); a full verification gate; and the powered user study the month cannot afford.

4.2 The chosen feature set as one vertical slice

The three features are not three projects; they are one workflow the receipt makes auditable: *a user-confirmed training-day target* → *an explained recommended meal* → *a pantry-aware missing-item To-Buy list*, on desktop and mobile. It maps onto the milestones above: N2 computes the line items, N3 renders the receipt and the one-tap “add missing to list” on the authenticated, responsive path, N4 tests the arithmetic and correctability, and N5 secures the list mutation (idempotent, no write before confirmation). **Primary one-month slice:** Feature 1 plus the narrow, user-confirmed Feature 3, with Feature 2 as an acceptance requirement (responsive + cross-session sync + WCAG status messages). **Deferred to Project 3:** native mobile apps, wearable/appliance integration, and any automatic workout-burn or weight-loss-rate estimation — each a safety- or scope-driven cut, not an oversight. Acceptance signals are pre-committed: 100% deterministic fixture agreement on pantry subtraction and target math; no list mutation before confirmation; exercise calories never double-counted; and $\geq 4/5$ task-test users able to say what will be added and why.

4.3 Support material we must read (excerpt)

80 laws, standards, licences and domain sources were mapped to use cases. Must-reads: WCAG 2.2 for the colour-only expiry

channel [17]; OWASP ASVS 5.0 for object-level authorisation [2] and LLM01 for the unbounded free-text goal interpolated into the Gemini prompt [4]; Supabase RLS [15]; FDA allergen guidance for recommendation wording [3]; and our own dependency licences (root `package.json`, `pyproject.toml`).

5 D5 — Prompt Report

5.1 Method

Every prompt went to a **fresh analyst with no memory** of any other, with an identical shared-evidence block pasted in, so agreements mean something. Four models were used: **Claude Opus 5** (42 web-grounded agents, 1,629 tool calls) and **OpenAI Codex** (conservative, single web-grounded run) did the primary survey; **DeepSeek V4-pro** and **Qwen2.5:32B** (run locally on 2×RTX A5000, *no* web retrieval) reran the twelve starters as third and fourth independent analysts.

5.2 Most and least useful prompts

Most useful: *C1 “catch them out”* (129 genuine errors, 2 dead products, 1 fabricated feature — highest value per token); *P17 disconfirmation* (the only prompt that made us lose, narrowing the central claim); *P07 the gap with receipts* (killed two gaps via a pre-committed found/not-found list). **Least useful:** *P08 mission* and *P05 stakeholders* — fluent, zero tool calls, unfalsifiable. The generalisation: prompts that can come back *against* us earned their marks; prompts that cannot be wrong in a checkable way produced our least defensible output.

5.3 P01 and P10 across models; the two-model rule

Table 6 contrasts the survey and red-team fronts. Six rivals clear the two-model bar (named by ≥ 2 LLMs); thirteen more clear the live-URL bar; three are struck as discontinued. Disagreement went both ways: Claude caught Codex carrying *Mealime* as live and resting the expiry-gap kill on a misread Eat This Much page; Codex reached the Samsung source Claude could not (HTTP 403).

5.4 Caught errors

Claude’s unverified first pass was wrong **129 times** (fabricated a product, *Fridgely*; quoted a marketing listicle as vendor feature text; carried two discontinued products as live). The catch that outranks them all was *our own*: a G1 confirming citation to a Cooklist review dated 2023-07-09 returned zero hits across 272 pulled reviews — **we withdrew it**. The offline models made the two-model rule earn its place: **Qwen2.5 fabricated URLs and prices wholesale**, exactly the failure the rule exists to filter, and **DeepSeek V4 spent its entire token budget on hidden reasoning**, returning an empty answer on one prompt — both are reportable properties, not usable evidence.

5.5 Model assessment, one line each

Claude Opus 5 — strength: sustained adversarial retrieval and reliable argument *against* its own priors; weakness: a confidently wrong first pass (129 errors). **Codex** — strength: conservative sourcing, explicit unknowns, measurable kill signals; weakness: single-model, small complaint sample, one stale rival. **DeepSeek V4-pro** — strength: strong structured reasoning when it finishes; weakness: no browsing via API, and its reasoning budget can starve the answer. **Qwen2.5:32B (local)** — strength: fast, private, zero marginal cost; weakness: with no retrieval it invents URLs and prices, which is precisely why it belongs behind the two-model rule, not in front of it.

5.6 Local model

A local model *was* run: Qwen2.5:32B via Ollama on the team’s dual-A5000 machine, all twelve starters, fresh context each. Its value was not its answers — it fabricated market evidence — but the demonstration that our team-level filter (two-model rule) catches an offline analyst that ignores the prompt-level evidence rules.

5.7 Disagreement as a deliverable, and the pivot

A dedicated prompt (C4) forced **twelve registered disagreements** between the Claude and Codex analysts, including one where Codex rejected git history as skill evidence and then used repository evidence to assert team capability three sections later. The most uncomfortable came from the pivot prompt (P12): given that the plan puts three things this team has *zero commits* behind on its critical path — a production `.tsx` change, a Supabase migration, and an RLS policy — the model recommended **pivot**, not stay-the-course. We keep that finding visible and unresolved rather than editing it away; it is the honest counterweight to a survey we otherwise “won.”

6 Threats to Validity

No user interviews and no rival installed by anyone on the team — the cheapest remaining improvement. Our surviving gap has *no demand-side evidence*. Verification reached 30 of 48 candidate products. Offline models cannot satisfy the live-URL rule and are reported as leads, not sources.

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Table 6: Cross-model comparison of the two rubric prompts.

	Claude Opus 5 (web)	Codex (web)	DeepSeek V4 (no web)	Qwen2.5:32B (local, no web)
P01 survey	14 live + 4 dead, adversarially fact-checked	7, refused to pad to 10	Terse; heavy internal reasoning, few names	10 named <i>with fabricated URLs and prices</i>
Evidence discipline	Live URL per claim, re-fetched twice	Live URL; explicit “unknown”	Recalled priors; cannot satisfy live-URL rule	Violated “quote-or-unknown”: invented \$9.99/mo cells, a dead product (Foodily), a non-product (“Fitbit Food”)
P10 red team	Grounded in Grocy’s formula, a null result, an HCI paper [16]	Grounded in vendor marketing	Plausible but source-free	Plausible but source-free

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